NLP Assignment 2 Group 6

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1 Introduction

In Assignment 1, a classical text classification pipeline was established for the AG News dataset using TF-IDF features and linear classifiers, with LinearSVC achieving a test Macro-F1 score of 0.9145. This assignment extends that work by replacing the linear classifier with two neural encoders that operate directly on token sequences: a Convolutional Neural Network (CNN) that captures local n-gram patterns through parallel convolution filters, and a Bidirectional Long Short-Term Memory network (BiLSTM) that models sequential context from both directions. All other conditions (dataset, splits, tokenizer, and evaluation metrics) are kept identical to Assignment 1 to allow a direct comparison. A small ablation study is also conducted to isolate the effect of dropout regularization on generalization.

2 Dataset and Split Details

The dataset, preprocessing steps, and split configuration are identical to Assignment 1. The AG News dataset provides a predefined split of 120,000 training instances and 7,600 test instances. A validation set was carved out from the training data using a 90/10 split with random seed 36. This yielded 108,000 training instances and 12,000 validation ones. The test set is evaluated exactly once, after model selection, on the validation set.

3 Tokenization and Vocabulary

Tokenization uses a regex tokenizer (similar to Assignment 1) to split the text into words, and the sequences are converted to integer indices for use with an embedding layer. Vocabulary is built from the training set, keeping only words that appear more than twice and reserving two special tokens: ¡pad¿ (index 0) for padding sequences and ¡unk¿ (index 1) for out-of-vocabulary words.

The vocabulary contains approximately 64,000 tokens. Sequences are truncated or padded to a maximum length of 64 tokens, with padding tokens added to sequences shorter than 64 tokens.

4 Model Architectures

Both models use an `nn.Embedding` layer of 100 dimensions initialized randomly, and are trained under identical conditions, summarised in

Table 1: Shared training hyperparameters for both models.

Hyperparameter	**Value**
Embedding dimension	100
Maximum sequence length	64 tokens
Batch size	64
Optimizer	Adam ($lr = 10^{-3}$)
Loss function	Cross-entropy
Early stopping patience	5 epochs (monitored on val Macro-F1)
Maximum epochs	50
Random seed	36

Table 2: Architecture summary.

Component	**CNN**	**BiLSTM**
Encoder	3× Conv1d (k=3,4,5) + ReLU + MaxPool	BiLSTM (hidden=64/direction)
Representation dim	300 (100 × 3 kernels)	128 (64 × 2 directions)
Dropout (baseline)	0.00	0.00
Output layer	Linear (300 → 4)	Linear (128 → 4)

4.1 CNN Text Classifier

The CNN follows the TextCNN architecture. After embedding, the sequence tensor is transported to shape (*batch*, *embed_dim*, *seq_len*) and passed through three parallel `nn.Conv1d` layers with kernel sizes 3, 4, and 5, each producing 100 feature maps. Each output is activated with ReLU and global max-pooled over the time dimension, yielding one scalar per feature map. The three pooled vectors are concatenated into a 300-dimensional representation, which passes through a dropout layer and a linear output layer producing four logits. The design captures local n-gram patterns at three different window sizes simultaneously.

4.2 BiLSTM Text Classifier

The BiLSTM encoder uses a single-layer bidirectional LSTM with hidden dimension 64 per direction. The final hidden states from the forward and backward

passes are concatenated to form a 128-dimensional sentence representation, which passes through a linear output layer. This architecture captures long-range dependencies and attends to context from both directions of the input.

5 Initial Results

5.1 CNN Text Classifier

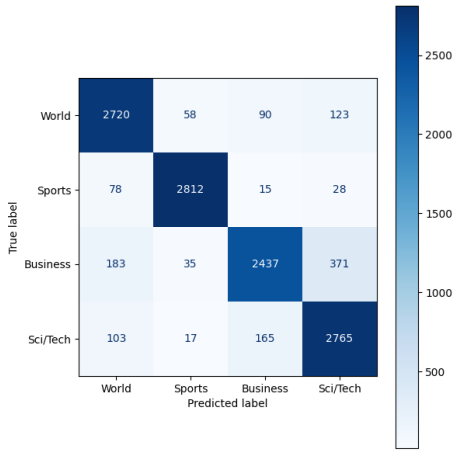

Figure 1: CNN Confusion Matrix

Metric	Value
Val Loss	0.6432
Val Accuracy	0.8945
Val Macro-F1	0.8947

Figure 2: CNN Validation Results

5.2 BiLSTM Text Classifier

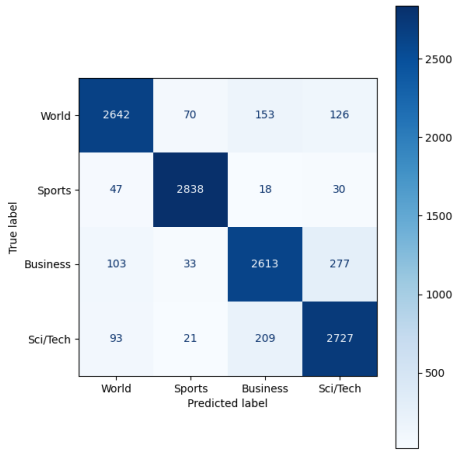

Figure 3: BiLSTM Confusion Matrix

Metric	Value
Val Loss	0.3966
Val Accuracy	0.9017
Val Macro-F1	0.9022

Figure 4: BiLSTM Validation Results

5.3 Learning Curves

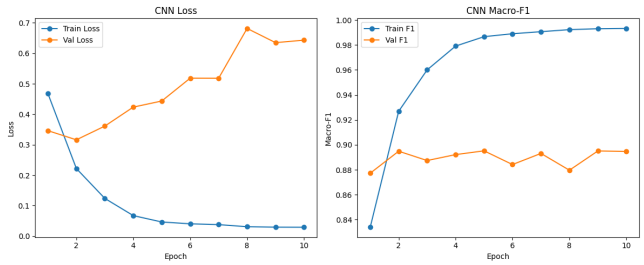

Figure 5: CNN Learning Curves

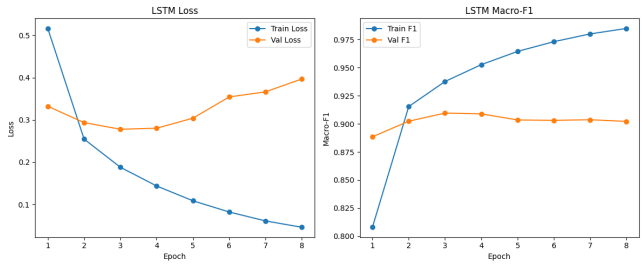

Figure 6: BiLSTM Learning Curves

Both models show signs of overfitting. The CNN has a rapidly decreasing training loss, but the validation loss starts increasing after a few epochs, and the validation F1 plateaus. The BiLSTM shows a similar trend, with validation loss diverging from the training loss, indicating the models are overfitting to the training data and struggling to generalize well.

6 Ablation

To combat this overfitting, we added a dropout rate of 0.3 to both models and re-evaluated their performance. This regularization technique helps prevent the models from becoming overly reliant on specific features during training, improving their generalization ability on unseen data. The new values are shown in the table below.

Metric	CNN	BiLSTM
Validation Loss	0.5149	0.3481
Validation Accuracy	0.9028	0.9126
Validation Macro-F1	0.9033	0.9130

Table 3: CNN and BiLSTM Validation Results after Dropout Regularization

7 Testing

After applying a dropout rate of 0.3 to both the CNN and BiLSTM models, the BiLSTM with dropout emerged as the best performing model in the validation phase, as it dealt with the overfitting problem. Consequently, this model was selected for testing on the holdout set. The following table displays the results for the BiLSTM model, which was evaluated on the holdout set, showing its performance in terms of loss, accuracy, and Macro-F1 score, as well as it's confusion matrix.

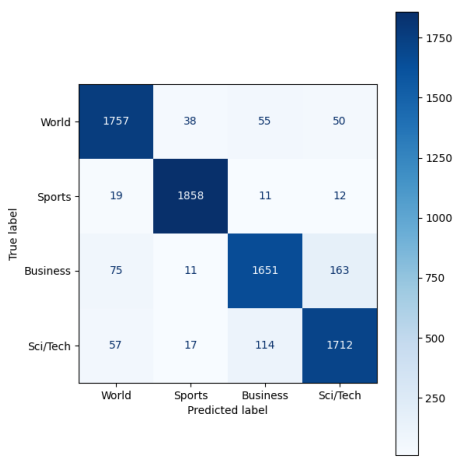

Figure 7: Testing Confusion Matrix

Metric	BiLSTM (Test)
Test Loss	0.3224
Test Accuracy	0.9182
Test Macro-F1	0.9180

Figure 8: BiLSTM Test Results

8 Error Analysis

The BiLSTM made 622 errors on the 7,600 test instances (overall error rate: 8.2%), compared to 648 for Assignment 1. The same three confusion patterns from Assignment 1 still persist: Business ↔ Sci/Tech remaining the dominant source of errors, driven by technology company articles that mix financial and technical vocabulary. World ↔ Business confusion arises in articles about economic policy and trade agreements, where geopolitical and financial language both occur. Sports is the cleanest class, though a small number of articles mentioning athletes or teams in non-sports contexts still get misclassified.

Table 4 compares the full error profile against Assignment 1. The largest improvements are on World → Business (−28) and World → Sports (−18), where sequential context helps the model identify the primary topic rather than being anchored to individual keywords. Business → World (+16) and Business → Sci/Tech (+8) increased slightly, suggesting the BiLSTM is occasionally over-sensitive to financial language appearing in non-Business articles.

Table 4: Error profile comparison: Assignment 1 (LinearSVC) vs. Assignment 2 (BiLSTM)

Confusion pair	**A1**	**A2**	**Δ**
Business → Sci/Tech	155	163	+8
Sci/Tech → Business	129	114	−15
World → Business	83	55	−28
Business → World	59	75	+16
World → Sports	56	38	−18
Sci/Tech → World	52	57	+5
World → Sci/Tech	45	50	+5
Sports → World	10	19	+9
Total errors	648	622	−26
Test Macro-F1	0.9145	0.9180	+0.0035

On the following table we analyze 10 examples of misclassified articles.

True Label	**Predicted Label**	**Sample**	**Analysis**
Business	Sci/Tech	"Card fraud unit nets 36,000 cards in its first two years, the UK's dedicated card fraud unit has recovered 36,000 stolen cards and 171 arrests, and estimates it saved £65 million."	The article discusses fraud prevention in business operations but is misclassified as Sci/Tech, likely due to its focus on technology used for fraud detection.
Business	Sci/Tech	"IBM to hire even more new workers by the end of the year, the computing giant plans to have its biggest headcount since 1991."	While IBM is a tech company, the focus of the article is on workforce and business operations. The model mistakenly categorized it as Sci/Tech due to the mention of IBM.

True Label	Predicted Label	Sample	Analysis
Business	Sci/Tech	"Rivals try to turn tables on Charles Schwab by Michael Liedtke – San Francisco (AP) – With its low prices and iconoclastic attitude, discount stock broker Charles Schwab Corp. (SCH) represented an announcement."	This article discusses the stock market and business competition but was misclassified as Sci/Tech. The overlap between technology and business with the stock market mention led to the confusion.
World	Sci/Tech	"Venezuela prepares for Chavez recall vote supporters and rivals warn of possible fraud; government says Chavez's defeat could produce turmoil in world oil market."	The article is about politics and fraud in the context of a Venezuelan election. The model misclassified it as Sci/Tech because of the mention of fraud, which could have been confused with technology-related fraud.
Business	World	"Promoting a shared vision as Michael Kaleko kept running into people who were getting older and having more vision problems, he realized he could do something about it."	The article, which discusses a vision-related business venture, was misclassified as World news, likely because the focus was on personal health and not explicitly on business.
World	Sports	"Olympics Day Four Richard Faulds and Stephen Parry are going for gold for Great Britain on Day Four in Athens."	The article should have been classified under World news, but the model misclassified it as Sports, likely due to the Olympics being a global event that covers multiple categories.
Sci/Tech	Business	"Intel to delay product aimed for high-definition TVs San Francisco – in the latest of a series of product delays, Intel Corp. has postponed the launch of a video display chip it had previously planned to release."	Although this article discusses a technology delay by Intel, the model classified it as Business, as it seems to focus on the business aspects of the article.
Sports	World	"Kederis proclaims innocence Olympic champion Kostas Kederis today left hospital ahead of his date with IOC inquisitors claiming his innocence and vowing: 'After the crucifixion comes the resurrection.'"	This article focuses on sports and the Olympics, but the model mistakenly classified it under World news. The mention of the IOC and inquisitors might have confused the model.

True Label	Predicted Label	Sample	Analysis
Sci/Tech	Business	"Yahoo! Ups Ante for Small Businesses Web giant Yahoo! is gambling that price cuts on its domain name registration and web hosting products will make it more competitive with discounters in the space."	This article discusses Yahoo's business strategy in the web hosting and domain registration industry. Given the business nature of the article, it was misclassified.
Sci/Tech	World	"Oil prices bubble to record high the price of oil has continued its sharp rise overnight, closing at a record high."	This article about oil prices is labeled as Sci/Tech, but the model misclassified it as World news due to the global nature of oil prices.

9 Reproducibility Notes

All experiments were run on Google Colab (Python 3.12). The full source code, notebook, and environment file are available at:

`https://github.com/BogdanI40/NLP_Group6`

The repository has the following structure:

```
Assignment2/
  README.md            <- run instructions and results summary
  requirements.txt     <- exact package versions
  assignment2.ipynb    <- full results
  assignment2.py    <- full pipeline (data -> features -> model -> eval)
  report/
    NLP_Group6.pdf  <- report
```

1. Install dependencies.

```
pip install -r requirements.txt
```

2. Run the code.

```
python assignment2.py
```

The code automatically downloads the AG News dataset from HuggingFace, performs preprocessing, trains both models, and outputs all metrics and figures. Expected runtime: approximately 3–4 hours on a standard CPU.

3. Visualization

```
python assignment2.ipynb
```

As the code takes a long time to run, assignment2.ipynb is also available as the results are already displayed there.

Table 6: Reproducibility safeguards.

Safeguard	**Detail**
Random seed	`SEED = 36` used in `train_test_split` and `LinearSVC(random_state=SEED)`
Vectoriser fit	`TfidfVectorizer` fitted on train only; dev and test are transformed only
Hyperparameter search	5-fold CV on train set only; dev used solely for final model selection
Test set policy	Test evaluated exactly once, after model selected on dev set
Dataset source	HuggingFace: `sh0416/ag_news` (official AG News predefined split)

Key reproducibility safeguards.